

Bronchiolitis

Contents

Bronchiolitis	1
Outline	1
Introduction	1
Epidemiology	2
Predisposing factors	2
Aetiology	2
Pathogenesis	2
Host-related factors	3
Immunological factors	3
Microbiological factors	3
Pathology	3
Pathophysiology	3
Clinical features	4
Complications and co-morbidities	4
Investigations	4
Differential diagnosis	4
Treatment	5
Prevention and control	5
References	5

Bronchiolitis

Dr Odochi Ewurum, MBBS, FWACP, FMCPaed – Paediatrics, topic 38

Outline

Introduction · Epidemiology & predisposing factors · Aetiology · Pathogenesis · Pathology · Pathophysiology · Clinical features · Complications · Investigations · Differential diagnosis · Treatment · Prevention

Introduction

Bronchiolitis is an inflammatory obstruction of the small airways – the bronchioles and surrounding tissues.

- It is an **acute infection of the lower respiratory tract**
- **Predominantly a viral disease**, with few cases attributed to other organisms

Epidemiology

- Peak incidence during the **rainy season** in tropical countries (winter in temperate countries)
- Prevalence is **higher in urban areas**
- Occurs during the **first 2 years of life**
- **Peak age 2-8 months (average 6 months)**
- **Commoner in males – M:F 2:1**
- Most cases are **mild, lasting 7-10 days**
- **Low socio-economic status** increases the risk of hospitalization
- **Mortality** in hospitalized cases is about **1-2%**, occurring mostly in children **< 6 months**; higher in developing countries due to poor nutrition and healthcare

Predisposing factors

1. Rainy season
2. Cold, dry, dusty harmattan season
3. Babies who are **not breastfed**
4. **Overcrowding**, day-care centres
5. Exposure to **cigarette smoke**
6. **Low birth weight**
7. **Chronic lung disease**
8. **Congenital heart lesions**
9. **Older family members** as a common source of infection

Aetiology

Viruses (predominant):

- **Respiratory syncytial virus (RSV)** – the commonest
- Adenovirus
- Parainfluenza 1, 2, 3
- Human metapneumovirus
- Bocavirus
- Rhinovirus
- Influenza virus

Bacteria (few cases):

- *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, *Simkania negevensis*, *Staphylococcus aureus*, *Staph epidermidis*, *Klebsiella pneumoniae*

Pathogenesis

Acquisition occurs through:

- **Inhalation** of infected respiratory droplets
- **Inoculation** of the nostrils/conjunctivae from contaminated hands of contacts

The initial viral infection and replication occurs in the **infant's upper respiratory tract**, then spreads down.

Several **risk factors** determine host vulnerability, infectivity and severity – **host**, **immunological**, **environmental** factors, and **microbiological characteristics** of the pathogen.

Host-related factors

- **Age-related peripheral airway dimension – the dominant factor.** The innately **smaller peripheral airways** of infants have a greater propensity for **severe obstruction**
- **Paucity of avenues for collateral ventilation** (e.g. pores of Kohn, canals of Lambert)
- **Comparatively greater number of mucus-secreting (goblet) cells** in infants
- **Genetic age-related airway hyper-reactivity** to RSV and parainfluenza type 3

Immunological factors

RSV incites complex immune responses:

- **Degranulation of eosinophils** and release of **eosinophil cationic protein** (cytotoxic to airway epithelium)
- Increased **Toll-like receptor-4** expression
- Increased lymphokines, chemokines, cytokines, **IgE antibody** and inflammatory mediators – **IL-8, MIP-1, RANTES, monocyte chemotactic protein-1, neutrophil chemotactic factors, interferon- γ , leukotriene C4, virus-specific IgE, IgG-4, histamine and eosinophilic cationic protein**

Microbiological factors

- **RSV group A** is associated with **greater severity** than group B
- **Parainfluenza type 3** is frequently associated with severity in blacks
- **Adenovirus types 7, 11, 21** are associated with severe bronchiolitis, more likely complicated by **bronchiolitis obliterans**

Pathology

- **Necrosis of the respiratory epithelium** and destruction of ciliated cells
- **Intraluminal bronchiolar plugs** of sloughed cellular debris and fibrin
- **Peribronchial infiltration** by mononuclear cells
- Extensive epithelial destruction, **pulmonary parenchymal necrosis** and **hyaline membrane formation** – all **more in adenovirus** than RSV
- A **profligate cellular and humoral immune response** with cellular injury – in RSV

Recovery: epithelial regeneration begins about **4 days** after the acute phase; **ciliary activity** is restored in **~2 weeks or more**; sloughed plugs are removed by scavenging **macrophages**.

Pathophysiology

Oedema of the bronchiolar mucosa + mucus and cellular debris + muscle constriction -> reduced lumen size -> increased resistance in the small airways during **both inspiration and expiration**.

Because resistance rises **exponentially (four-fold)** as airway radius falls, the **work of breathing, trapped (dead-space) air and respiratory rate** all increase.

- The airway radius is **smaller during expiration**, so obstruction causes **early air trapping** beyond the site and **progressive hyperinflation** – a **ball-valve** obstruction
- If obstruction becomes **complete (check-valve)** -> **resorption of trapped distal air -> patchy atelectasis**
- **Hypoxaemia** results from **ventilation-perfusion mismatch** early in the disease

- With severe obstruction and tiring effort, less air reaches the alveoli -> **hypercapnia** (raised PCO₂) may develop

Clinical features

Symptoms: sneezing and clear rhinorrhoea; diminished appetite; fever **38.5-39°C** (or subnormal temperature); respiratory distress; **paroxysmal wheezy cough**; dyspnoea; irritability; **tachypnoea**; **apnoea**.

Signs:

- **Markedly increased work of breathing** – nasal flaring, retractions
- **Fine inspiratory crepitations and rhonchi**
- **Prolonged expiratory phase**
- **Reduced breath sounds**
- **Hepatomegaly and splenomegaly** – due to **lung hyperinflation** pushing the organs down (not true organomegaly)

Complications and co-morbidities

- **Dehydration, apnoea and bradycardia** – especially in infants < **2 months**
- Acute otitis media
- **Hypoxaemia** ± hypercarbia
- Viral pneumonia; secondary bacterial pneumonia
- **Recurrent wheezing or asthma**
- Urinary tract infection

Investigations

Chest X-ray (PA and lateral):

- May be **normal**
- **Peribronchial thickening**
- **Hyperinflation** – “splaying of the ribs”, flattened diaphragmatic domes, **relative microcardia**
- **Patchy atelectasis**, increased bronchovascular markings, interstitial streaks from the hilum
- Increased **antero-posterior diameter** on lateral film

FBC – total WBC and differentials usually **normal**.

Viral testing (rapid immunofluorescence, PCR, or culture) – if diagnosis is uncertain or for epidemiology.

Pulse oximetry – detect hypoxaemia and guide oxygen need.

Blood gas/acid-base – for aberrations like metabolic acidosis.

Differential diagnosis

Bronchial asthma; wheeze-associated respiratory infection (WARI); congestive cardiac failure; pertussis; aspirated foreign body; gastro-oesophageal reflux disease; croup; retropharyngeal abscess; acute epiglottitis; impacted foreign body in the upper airways.

Treatment

Usually supportive. Current recommendations:

- **Oxygen therapy**
- **Increased fluids and feeds**
- **Trial of bronchodilators**
- **Corticosteroids / anti-inflammatory agents** (e.g. leukotriene) – only if benefit outweighs risk
- **Antiviral (ribavirin)** – delivered by a **small-particle aerosol generator (SPAG)** via an oxygen hood/ventilator, over 3-5 days (18-20 h/day); reserved for **severe cases, the immuno-compromised, or infants with cyanotic CHD**
- **Antibiotics** – only with concomitant severe bacterial infection
- **Physiotherapy** – for secretion clearance; not routine
- **Antipyretics**, nasal decongestion, cough mixtures

Prevention and control

- **Palivizumab immunoprophylaxis** (intramuscular)
- **Health education on hand decontamination** – health workers, parents, nursery/creche staff; attention to inter-child hand-washing
- **Encouraging breastfeeding**
- Minimizing spread in creches/nurseries by excusing children and staff with acute respiratory illness
- Discouraging **cigarette smoking and domestic smoke exposure**

References

Watts KD, Goodman DM. Wheezing, Bronchiolitis and Bronchitis. In: *Nelson's Textbook of Paediatrics*, 19th ed. Elsevier; 2011. Johnson ABR et al. Pneumonias and other ALRI. In: Azubuike JC, Nkanginieme KEO. *Paediatrics and Child Health in a Tropical Region*, 3rd ed.